

Client Onboarding for Oil Commander REST API

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REST API Usage Guide (Business & Technical Audience)

Overview

This document explains how to access and use the **Oil Commander REST API** using **OAuth 2.0 Client Credentials**.

It is designed for:

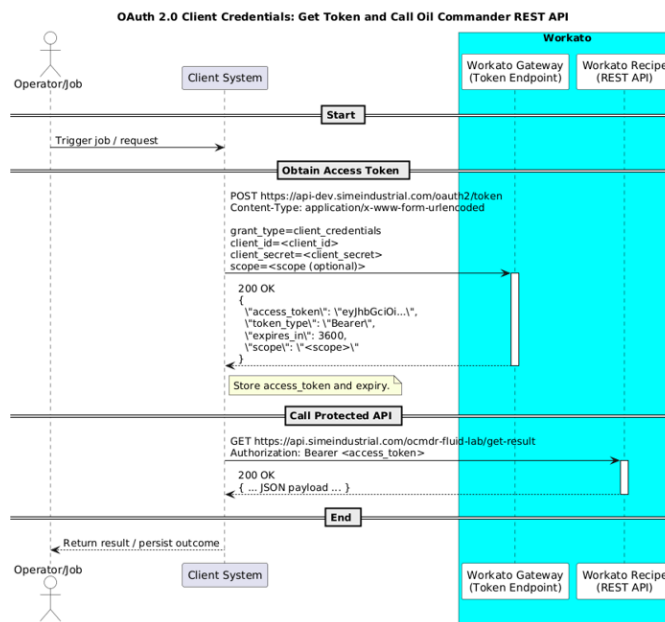
- **Business users** who need to understand what the API does and how it fits into a workflow.
- **Technical users** who need exact details on authentication, endpoints, and request/response formats.

What This API Does

The Oil Commander API allows external systems to:

- Request fluid analysis results.
- Retrieve operational data for integration into business processes.

Abstract Usage Flow



Abstract Token Flow

Technical Details

Step 1 - Obtain Access Token

This API is configured to use OAuth 2.0 Client Credentials Grant Authentication mechanism.

This requires the calling client to get the **Bearer** token from the Workato Identity Server using the following CURL Specification.

Token Request

```
1 curl --location 'https://api.simeindustrial.com/oauth2/token' \
2 --header 'Content-Type: application/x-www-form-urlencoded' \
3 --data-urlencode 'grant_type=client_credentials' \
4 --data-urlencode 'client_id={Client-Id}' \
5 --data-urlencode 'client_secret={Client-Secret}'
```

Note:

- Please substitute Client Id and Client Secret in the below curl request that you got from the Enterprise Automation Team.

Token Response

```
1 {
2   "access_token": "eyJhbGciOi...",
3   "token_type": "Bearer",
4   "expires_in": 3600,
5 }
```

Note:

- The Token is valid for 1 hr only. Please configure the Caching of the token on client side to refresh every hour or so.

Step 2 - REST API Usage Samples

There are two environment hostnames for this API

- Workato **PROD**: `https://api.simeindustrial.com/ocmdr-fluid-lab/get-result`
- Workato **TEST**: `https://api-test.simeindustrial.com/ocmdr-fluid-lab/get-result`

API Request

```
1 curl --location 'https://{Environment-Hostname}/ocmdr-fluid-lab/get-result?start_date=2019-02-
12&start_time=00%3A00%3A00.000&end_date=2019-02-13&end_time=23%3A59%3A59.000' \
2 --header 'user-email: {Email-Address}' \
3 --header 'Accept: application/json' \
4 --header 'Authorization: Bearer {Token}'
```

Note:

- Please substitute {Token} in the above curl request with the token received in Step 1.
- Please substitute {Environment Hostname} for appropriate environment you want to Integrate with.
- Please substitute {Email-Address} with the registered Oil Commander email address.
- Please substitute appropriate values for the Start & End date/time.
- If you do not pass Start and End times, they will be defaulted with start time as 00:00:00.000 and end time as 23:59:59.999.
- The API supports “application/json” and “application/xml” data formats.

REST API Headers and Parameters explained

Headers

Name	Mandatory (M)/ Non-Mandatory (NM)	Comments/Defaults
Authorization	M	Bearer Token fetched from Workato for Authentication
user-email	M	xxxx@something.com - a valid Oil Commander registered email address

Query Params

Name	Mandatory (M)/ Non-Mandatory (NM)	Comments/Defaults/Sample Data
------	-----------------------------------	-------------------------------

customer_name	NM	MAJOR CUSTOMER
jobsite_name	NM	TRUCKS
equipment_serial_number	NM	CAT1231231
start_date (yyyy-mm-dd)	NM	2024-01-01
start_time (hh:mm:ss.fff)	NM	08:00:00.000 (Defaults to 00:00:00.000)
end_date (yyyy-mm-dd)	NM	2024-01-01
end_time (hh:mm:ss.fff)	NM	16:00:00.000 (Defaults to 23:59:59.999)

Note:

- If no parameters are passed to the rest api, the default result would be to fetch 24 hours of data for the current date for all the associated customers to the {user-email} header

REST API Sample Response Payload

Sample Payload - 200

```
1  [
2    {
3      "customer_name": "XXX",
4      "customer_id": "XXX",
5      "jobsites": [
6        {
7          "name": "XXX",
8          "region": "XXX",
9          "equipment": [
10             {
11               "model_name": "XXX",
12               "make_name": "XXX",
13               "serial_no": "XXX",
14               "unit_no": "XXX",
15               "rego_no": "XXX",
16               "market_status": "XXX",
17               "compartments": [
18                 {
19                   "fluid_change_interval": 250,
20                   "compartment_id": "XXX",
21                   "fluid_sample_interval": 250,
22                   "compartment_name": "XXX",
23                   "make_name": "XXX",
24                   "model_name": "XXX",
25                   "sump_capacity": 10,
26                   "serial_no": "XXX",
27                   "samples": [
28                     {
29                       "fluid_added": "XXX",
30                       "sample_id": "XXX",
31                       "oil_type": "XXX",
32                       "sample_type": "XXX",
33                       "confirmed": "XXX",
34                       "fmu": "XXX",
35                       "oil_grade": "XXX",
36                       "evaluation": "XXX",
37                       "filter_changed": "XXX",
38                       "smu": 88413,
39                       "cmu": "XXX",
40                       "date_received": "XXX",
41                       "date_first_processed": "XXX",
42                       "fuel_type": "XXX",
43                       "fluid_changed": "XXX",
44                       "date_sampled": "XXX",
45                       "sample_label": "XXX",
46                       "customer_notes": "XXX",
47                       "coolant_type": "XXX",
48                       "date_modified": "XXX",
49                       "interpreter": "XXX",
50                       "interpretation": "XXX",
51                       "jobno": "XXX",
52                       "debris_stage": "XXX",
53                       "sampled_site": "XXX",
54                       "date_last_processed": "XXX",
55                       "actioned": 0,
```

```
56 "resolved": "XXX",
57 "actionsTaken": "XXX",
58 "analysis": [
59   {
60     "unique_id": 18,
61     "analysis_id": "XXX",
62     "test": [
63       {
64         "unique_id": 4,
65         "element_id": "XXX",
66         "result": "XXX"
67       },
68       {
69         "unique_id": 5,
70         "element_id": "XXX",
71         "result": "XXX"
72       },
73       {
74         "unique_id": 11,
75         "element_id": "XXX",
76         "result": "XXX"
77       },
78       {
79         "unique_id": 12,
80         "element_id": "XXX",
81         "result": "XXX"
82       },
83       {
84         "unique_id": 13,
85         "element_id": "XXX",
86         "result": "XXX"
87       },
88       {
89         "unique_id": 14,
90         "element_id": "XXX",
91         "result": "XXX"
92       },
93       {
94         "unique_id": 23,
95         "element_id": "XXX",
96         "result": "XXX"
97       },
98       {
99         "unique_id": 25,
100        "element_id": "XXX",
101        "result": "XXX"
102      },
103      {
104        "unique_id": 39,
105        "element_id": "XXX",
106        "result": "XXX"
107      },
108      {
109        "unique_id": 49,
110        "element_id": "XXX",
111        "result": "XXX"
112      },
113      {
114        "unique_id": 51,
115        "element_id": "XXX",
116        "result": "XXX"
117      },
118      {
119        "unique_id": 54,
120        "element_id": "XXX",
121        "result": "XXX"
122      },
123      {
124        "unique_id": 56,
125        "element_id": "XXX",
126        "result": "XXX"
127      },
128      {
129        "unique_id": 59,
130        "element_id": "XXX",
131        "result": "XXX"
132      },
133      {
134        "unique_id": 61,
135        "element_id": "XXX",
136        "result": "XXX"
137      },
138      {
139        "unique_id": 75,
140        "element_id": "XXX",
141        "result": "XXX"
142      },
143      {
144        "unique_id": 80,
145        "element_id": "XXX",
146        "result": "XXX"
147      },
148      {
149        "unique_id": 101,
150        "element_id": "XXX",
```

```

151         "result": "XXX"
152     },
153     {
154         "unique_id": 102,
155         "element_id": "XXX",
156         "result": "XXX"
157     },
158     {
159         "unique_id": 103,
160         "element_id": "XXX",
161         "result": "XXX"
162     },
163     {
164         "unique_id": 106,
165         "element_id": "XXX",
166         "result": "XXX"
167     },
168     {
169         "unique_id": 127,
170         "element_id": "XXX",
171         "result": "XXX"
172     },
173     {
174         "unique_id": 129,
175         "element_id": "XXX",
176         "result": "XXX"
177     },
178     {
179         "unique_id": 140,
180         "element_id": "XXX",
181         "result": "XXX"
182     },
183     {
184         "unique_id": 149,
185         "element_id": "XXX",
186         "result": "XXX"
187     },
188     {
189         "unique_id": 223,
190         "element_id": "XXX",
191         "result": "XXX"
192     },
193     {
194         "unique_id": 224,
195         "element_id": "XXX",
196         "result": "XXX"
197     }
198 ]
199 },
200 {
201     "unique_id": 43,
202     "analysis_id": "XXX",
203     "test": [
204         {
205             "unique_id": 32,
206             "element_id": "XXX",
207             "result": "XXX"
208         }
209     ]
210 },
211 {
212     "unique_id": 26,
213     "analysis_id": "XXX",
214     "test": [
215         {
216             "unique_id": 47,
217             "element_id": "XXX",
218             "result": "XXX"
219         },
220         {
221             "unique_id": 193,
222             "element_id": "XXX",
223             "result": "XXX"
224         },
225         {
226             "unique_id": 194,
227             "element_id": "XXX",
228             "result": "XXX"
229         },
230         {
231             "unique_id": 192,
232             "element_id": "XXX",
233             "result": "XXX"
234         }
235     ]
236 },
237 {
238     "unique_id": 24,
239     "analysis_id": "XXX",
240     "test": [
241         {
242             "unique_id": 82,
243             "element_id": "XXX",
244             "result": "XXX"
245         }
246     ]
247 },

```

```

246         {
247             "unique_id": 84,
248             "element_id": "XXX",
249             "result": "XXX"
250         },
251         {
252             "unique_id": 86,
253             "element_id": "XXX",
254             "result": "XXX"
255         },
256         {
257             "unique_id": 88,
258             "element_id": "XXX",
259             "result": "XXX"
260         },
261         {
262             "unique_id": 89,
263             "element_id": "XXX",
264             "result": "XXX"
265         },
266         {
267             "unique_id": 90,
268             "element_id": "XXX",
269             "result": "XXX"
270         },
271         {
272             "unique_id": 91,
273             "element_id": "XXX",
274             "result": "XXX"
275         },
276         {
277             "unique_id": 92,
278             "element_id": "XXX",
279             "result": "XXX"
280         }
281     ],
282     {
283         "unique_id": 38,
284         "analysis_id": "XXX",
285         "test": [
286             {
287                 "unique_id": 133,
288                 "element_id": "XXX",
289                 "result": "XXX"
290             }
291         ]
292     },
293     {
294         "unique_id": 29,
295         "analysis_id": "XXX",
296         "test": [
297             {
298                 "unique_id": 134,
299                 "element_id": "XXX",
300                 "result": "XXX"
301             }
302         ]
303     },
304     {
305         "unique_id": 25,
306         "analysis_id": "XXX",
307         "test": [
308             {
309                 "unique_id": 142,
310                 "element_id": "XXX",
311                 "result": "XXX"
312             }
313         ]
314     },
315     {
316         "unique_id": 15,
317         "analysis_id": "XXX",
318         "test": [
319             {
320                 "unique_id": 153,
321                 "element_id": "XXX",
322                 "result": "XXX"
323             },
324             {
325                 "unique_id": 155,
326                 "element_id": "XXX",
327                 "result": "XXX"
328             }
329         ]
330     }
331 ],
332 "image": []
333 }
334 }
335 ]
336 }
337 ]
338 }
339 ]
340 }

```

```
341     ]
342   }
343 ]
```

REST API Response Error Details

▼ Wrong email-id - Error Code = 400

```
1 {
2   "error": "Bad User",
3   "reason": "Please pass a valid user-email in the header",
4   "origin": "Header : user-email"
5 }
```

▼ When Start Date is greater than End Date - Error Code = 400

```
1 {
2   "error": "Bad Start Date or End Date",
3   "reason": "Start Date greater than End Date. Please correct the date range.",
4   "origin": "Parameter : start_date & end_date"
5 }
```

▼ Number of Days data Requested Exceeds 15 days - Error Code = 400

```
1 {
2   "error": "Bad Start Date or End Date",
3   "reason": "Max amount of days supported by this api is 15 days.",
4   "origin": "Parameter : start_date & end_date"
5 }
```

▼ When format of Start or End time is Wrong = Error-Code = 400

```
1 {
2   "error": "Bad Start time",
3   "reason": "Please pass valid start time in format - 00:00:00.000",
4   "origin": "start time"
5 }
```

▼ No Data Found - Error Code = 200

```
1 {
2   "query_parameters": {
3     "customerName": null,
4     "jobsiteName": null,
5     "equipmentSerialNum": null,
6     "startDate": null,
7     "startTime": null,
8     "endDate": null,
9     "endTime": null
10  },
11   "message": "NO_DATA_FOUND"
12 }
```

▼ When unexpected Error happens in API - Error Code = 500

```
1 {
2   "error": "Internal Server Error",
3   "reason": {Error message },
4   "origin": "Api Internal Error"
5 }
```